

Experiment Report: Pipelined FIR Filter Implementation and Resource Analysis

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1 Introduction

This report evaluates the performance and hardware resource utilization of Finite Impulse Response (FIR) filters implemented on the Intel Cyclone V FPGA (5CEBA2F17A7). We compare non-pipelined and pipelined architectures across three implementation methodologies: Direct, Optimized, and Genvar-based.

2 Pipelined Architecture Explanations

Pipelining increases maximum operating frequency (F_{max}) by inserting registers between combinational stages.

2.1 Direct Pipelined Form

The Direct Pipelined module (`dir_pipe.v`) registers multiplier outputs. However, the accumulation is performed combinatorially, creating a critical path in the adder tree, which limits F_{max} .

2.2 Optimized Pipelined Form

The Optimized module (`opt_pipe.v`) pipelines both multiplication and accumulation:

$$stage[i] \Leftarrow stage[i - 1] + ((x \cdot h[i]) \gg 14) \quad (1)$$

This reduces combinational depth to a single multiplier–adder path per stage.

2.3 Genvar Pipelined Form

The Genvar implementation (`genvar_pipe.v`) uses a Verilog `generate` loop to instantiate parameterized pipeline stages, enabling scalable hardware replication.

3 Analysis of Results

- **DSP Limitation:** Direct and unoptimized designs exceed available DSP resources, causing synthesis failure.
- **Register Overhead:** Pipelining increases register usage due to intermediate stage storage.

- **Efficiency:** Optimized and Genvar pipelined designs successfully fit within device constraints while maintaining throughput.

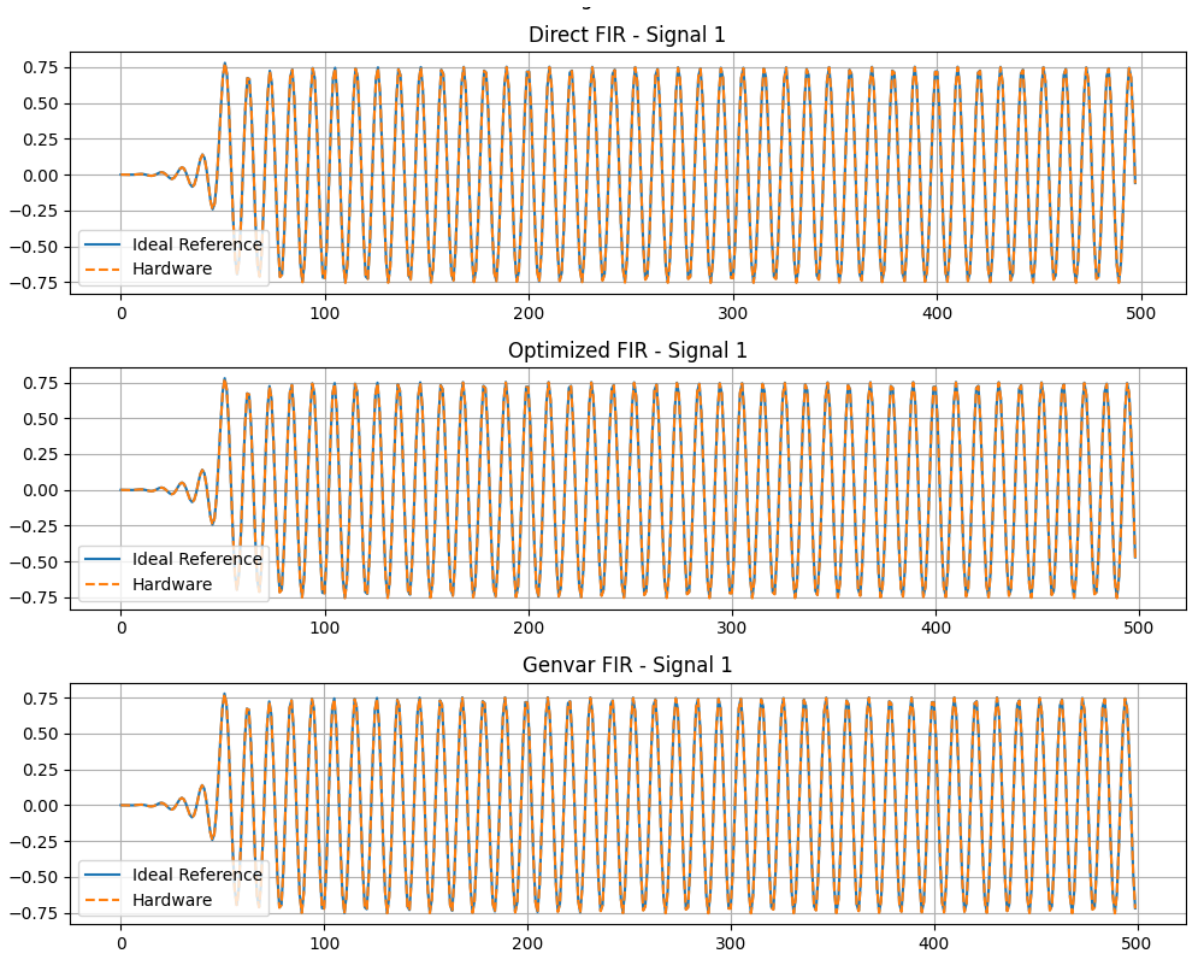


Figure 1: Signal 1 output comparison

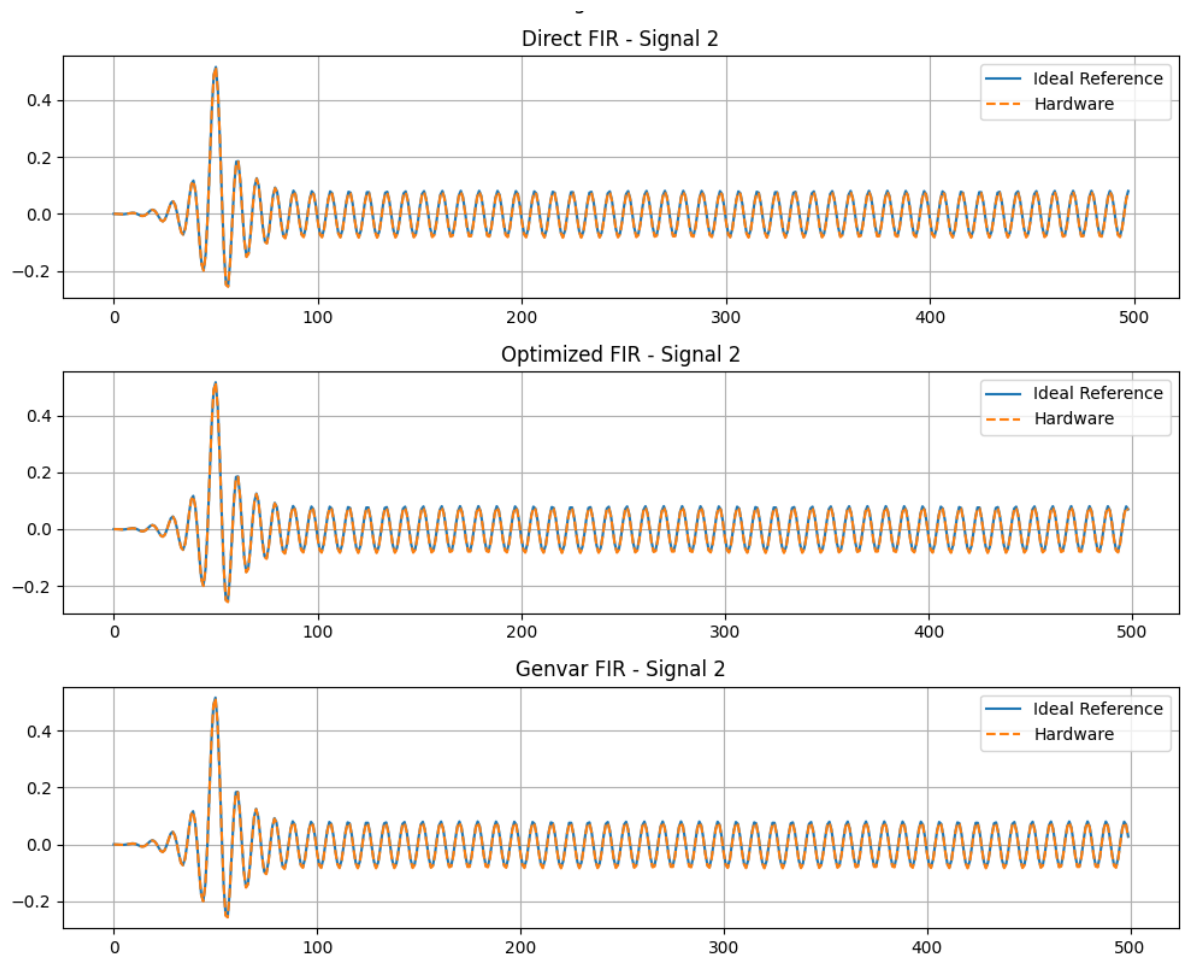


Figure 2: Signal 2 output comparison

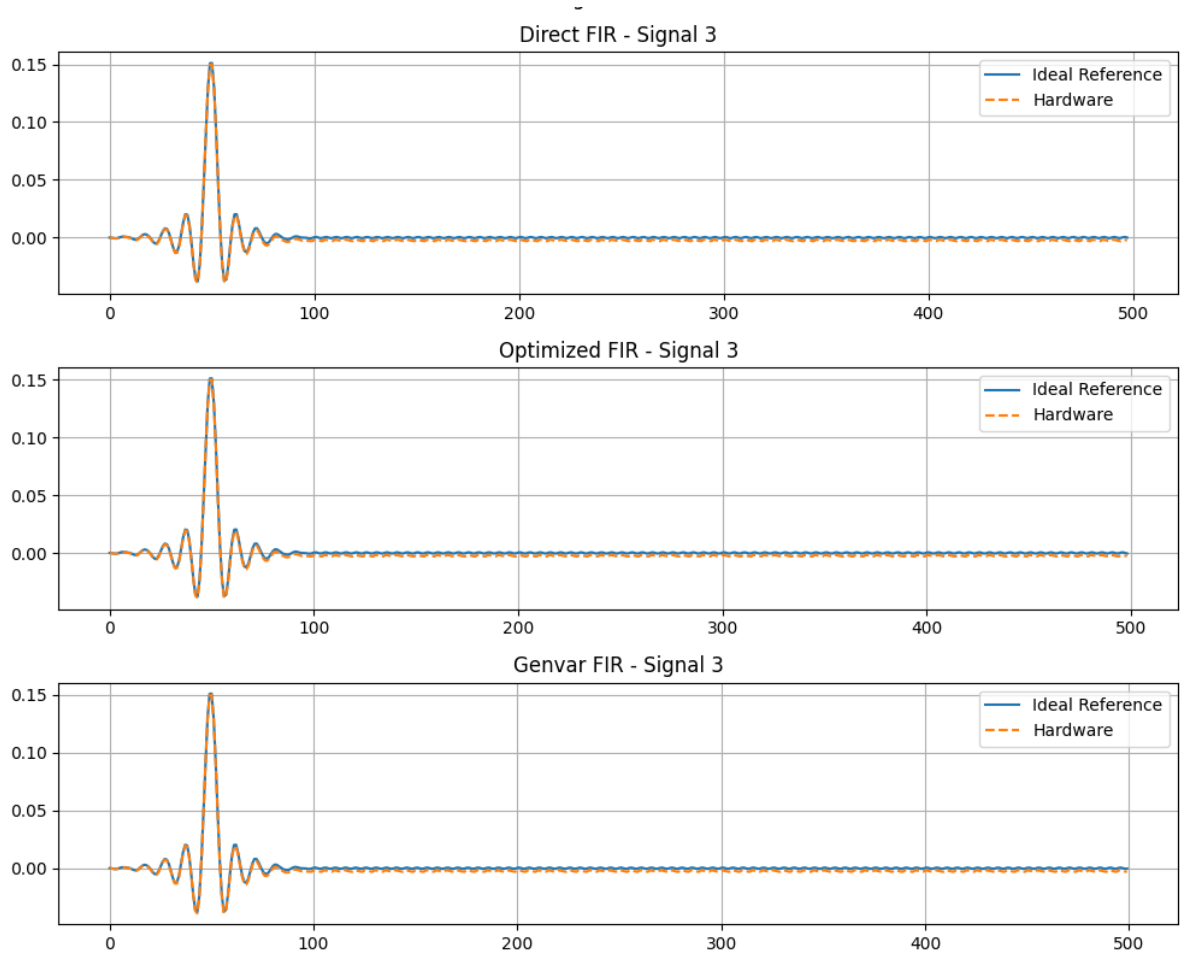


Figure 3: Signal 3 output comparison

4 Error Analysis and Performance Comparison

Signal	Method	Shift (cycles)	Max Error	Mean Error
Signal 1	Direct	2	0.0035306	0.0026868
	Optimized	1	0.0035306	0.0026872
	Genvar	1	0.0035306	0.0026869
Signal 2	Direct	3	0.0032893	0.0026387
	Optimized	2	0.0032893	0.0026391
	Genvar	1	0.0032893	0.0026397
Signal 3	Direct	3	0.0027478	0.0021957
	Optimized	2	0.0027478	0.0021961
	Genvar	1	0.0027478	0.0021972

Table 1: Comparison of error metrics and pipeline shift across FIR implementations

5 Comparison: User Modules vs. Intel FIR IP

The Intel FIR Compiler IP uses internal architectural optimizations such as pre-adders, dedicated pipeline registers, and DSP folding techniques, which are not directly exposed in RTL designs. This allows better resource utilization compared to manually written Verilog implementations.

6 Conclusion

Pipelining significantly improves timing closure in FIR filter implementations on FPGA devices. The optimized and genvar-based pipelined architectures achieve the best balance between DSP utilization, register overhead, and throughput, while direct architectures fail due to DSP resource exhaustion.